Industrial Control System (ICS) Analysis

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Introduction

Industrial Control Systems (ICSs) play a critical role in modern manufacturing because of their scalability, efficiency, and ability to automate processes in environments where continuous human presence is not always feasible. Toyota Motor Manufacturing, Kentucky (TMMK), relies on ICS technologies in the production of vehicles such as the RAV4 Hybrid. A significant portion of vehicle assembly and manufacturing is automated through Programmable Logic Controllers (PLCs), Distributed Control Systems (DCSs), industrial robots, and other control technologies. Additionally, complex software is integrated into vehicle systems to support safety and driver-assistance features such as lane detection and adaptive cruise control. Traditionally, ICS and operational technology (OT) environments have been designed primarily around availability, reliability, and continuous operation; as production interruptions can result in significant operational and financial consequences. Many of these systems were developed when OT networks were largely isolated or air-gapped, resulting in limited consideration of cybersecurity during their initial design (Srivastav et al. 2025).

The rapid development of data analytics, cloud computing, remote access, and other digital technologies has created significant opportunities for organizations to improve the efficiency and capabilities of OT environments. Connecting traditionally isolated OT systems to enterprise IT and business networks however, introduces cybersecurity risks that were not present in their original architectures. Unlike conventional IT systems, OT environments directly interact with physical equipment and processes, and a cyberattack can potentially affect not only the information systems but also manufacturing operations, safety, and production availability. This paper analyzes TMMK from the perspective of ICSs use within the modern automotive manufacturing environment and examines the cybersecurity challenges associated with protecting the OT infrastructure, and is organized around the ICS architecture, OT security issues, emerging trends, and the challenges associated with IT/OT convergence (Srivastav et al. 2025).

Architecture & System Monitoring

The Toyota manufacturing plant uses a segmented Purdue Model-based architecture to support Hybrid RAV4 production. The architecture separates Enterprise IT, operational technology (OT), supervisory systems, industrial controllers, and physical manufacturing equipment into distinct network environments. The primary structure is (*Project Architecture Document: Architectural Description of Toyota Motor Manufacturing, 2023*):

Enterprise IT → OT DMZ → Ops VLAN → Control VLAN → Floor VLAN

Enterprise IT provides corporate services such as ERP, authentication, DNS, file services, backups, patching, and firewall management. An Edge Firewall and Edge DMZ provide an additional protection from external networks. The R&D environment is separate from both Enterprise IT and OT and supports engineering and software development, including Safety Sense 2.0.

The OT DMZ serves as the primary security boundary between Enterprise IT and the Industrial network. It contains Supervisory Control and Data Acquisition (SCADA) servers and databases, control and historian servers, Automated Guided Vehicle (AGV) and inventory managers, patching services, read-only DNS/authentication, as well as security inspection tools. Internal firewalls restrict communication between the Enterprise, OT DMZ, and Industrial networks.

Within the Industrial environment, the Ops VLAN provides supervisory functions such as operator interfaces, monitoring, alarms, and scheduling. The Control VLAN contains PLCs/RTUs that execute control logic. The Floor VLAN contains field-level equipment, including robots, sensors, machinery, AGVs, environmental devices, and actuators. These supervisory systems issue authorized commands, controllers execute the logic, and field devices interact with the physical manufacturing process.

From a NIST SP 800-82 Rev. 2 perspective, the architecture demonstrates network segmentation, network segregation, DMZ-based boundary protection, controlled communication, and defense in depth. Multiple firewalls, VLANs, DMZs, and isolated networks work to limit unauthorized access and reduce the ability of a compromise in one environment to spread to critical manufacturing systems.

Cyber and Physical Security in Automotive OT

The architecture document names the Software Delivery/R&D system, which produces the safety-critical Toyota Safety Sense 2.0 software installed in each RAV4, as "one of the highest risk points" (TMMK, 2023, p. 9). R&D works on a physically isolated network, so once the software is tested and scanned, an employee carries it to the plant on an encrypted drive. Once on the plant side, it is sent wirelessly to the Trusted Platform Module (TPM) that goes into the vehicle and checked a final time by quality control after installation (TMMK, 2023, pp. 9–10) and the TPM gives the vehicle a hardware trust anchor for software updates (Plappert & Fuchs, 2023). The document is merely a summary, and the procedure may be stricter in practice; as written, the handoff itself has no equivalent safeguard; as nothing checks that the software on the drive is unchanged when it reaches the plant. While Quality control checks the software's hashes, a kind of digital fingerprint, but the document does not say what they are compared against. Best case scenario, a package altered at the handoff is caught once it is already in the car. The same drives also carry software to the OT DMZ cluster (TMMK, 2023, p. 13), home to the control and SCADA servers that supervise production, but the document describes no check on that path. As such, one compromised transfer could therefore reach both the vehicle and the ICS that builds it.

A further significant concern is the security of TMMK's industrial control systems (ICS), including Programmable Logic Controllers (PLCs), Distributed Control Systems (DCSs), engineering workstations, and industrial robots. Many OT systems were designed when manufacturing networks were largely isolated from IT environments; however, increasing IT/OT convergence has introduced additional attack surfaces and exposed legacy systems to new cybersecurity threats (Venanzi et al., 2026). A successful attack would potentially interrupt production or modify PLC logic, equipment configurations, or manufacturing parameters, resulting in defective end products. If this manipulation affects components associated with safety-related functions such as lane detection, adaptive cruise control, or radar systems, the consequences could extend beyond production losses to consumer safety risks, recalls, regulatory penalties, reputational damage, and manufacturer liability, resulting in human injuries and/or deaths. Cyber risks must also be considered in conjunction with physical threats from unauthorized individuals and disgruntled insiders who may have access to critical facilities, equipment, or systems. Consequently, TMMK requires a defense-in-depth strategy that integrates cybersecurity, OT security, TPM-based hardware security, supply-chain risk management, access controls, network segmentation,

monitoring, incident response, and physical security to protect both manufacturing operations and the safety of the vehicles produced.

Historical Threats and Cyber-Attack Vectors

Automotive and manufacturing supply chain attacks demonstrate how digital vulnerabilities manifest as physical operational crises; one such attack is the 2022 Kojima Industries ransomware attack. By encrypting key servers and severing the electronic data interchange (EDI) tied to Toyota's Just-In-Time (JIT) infrastructure, the malware forced the shutdown of 14 domestic manufacturing plants and 28 production lines in Japan for a day, halting the production of roughly 13,000 vehicles (Cullen, 2026). This operational disruption highlights broader systemic dependencies, mirroring industrial malware like Stuxnet, which manipulate internal loop networks and PLC ladder logic to override mechanical safety thresholds without alerting operators.

This logic is mirrored by the Toyota RAV4 Headlight CAN Bus Injection (CVE-2023-29389) exploit, in which vehicles automatically trust network messages without verification (CVE Details, 2023). Threat actors can bypass physical boundaries by pulling back the vehicle's front bumper or wheel well liner to access the exterior headlight wiring harness (Goodin, 2023). Attackers then splice a modified electronic injection device, such as a JBL Bluetooth speaker, directly into the exposed CAN bus wires (Goodin, 2023). The device then listens for the car to wake up, overrides the system by flooding the network with high-priority fraudulent messages to block legitimate ECUs from talking, and injects a forged "Key is validated" command (Halfacree, 2026). This allows adversaries to unlock the doors and steal the vehicle in under two minutes without a key fob (Car Expert, 2025).

These attacks succeed by exploiting unsegmented networks and missing flow control mechanisms within the automotive ecosystem (Goodin, 2023). The RAV4's architecture flaws place highly exposed exterior components on the same trusted network layer as safety critical body control modules (Goodin, 2023). This systemic lack of verification extends to external data channels, as demonstrated by a decade long Toyota cloud misconfiguration leak (2012–2023). Due to human error, public facing database settings exposed the data of over 2 million vehicle users, putting unique Vehicle Identification Numbers (VINs), navigation terminal locations, and onboard video footage at risk (Powell, 2026).

Evolution of ICS/OT Security

For several decades, industrial control system (ICS) security relied primarily on "security by obscurity." These systems operated using proprietary, unrouteable protocols such as Modbus, PROFINET, and EtherNet/IP, and networks that were physically isolated from information technology (IT) systems and the internet. The rationale was that if attackers could not access the network, traditional IT security measures were unnecessary. This assumption remained largely valid until manufacturers began connecting programmable logic controllers (PLCs) to Ethernet, implementing remote diagnostics, integrating supplier data feeds, and deploying over-the-air software updates. This transition, referred to as IT-OT convergence, is exemplified by TMMK's AGV system and Environmental SCADA, which use virtual local area networks (VLANs) and encrypted wireless communication rather than fully isolated hardware. While increased connectivity strengthens operational efficiency and visibility, it also expands the attack surface of networks not originally designed with authentication, encryption, or integrity verification.

Security engineering has evolved from single-point defense strategies, such as the air gap, to layered defense-in-depth approaches, and further toward resiliency engineering that presumes eventual breaches. Continuous monitoring with Intrusion Detection and Prevention Systems (IDPSs) examines Industrial Control System (ICS) network traffic for anomalous protocol behavior, as older protocols such as Modbus lack authentication. Active high-availability configurations ensure that critical control and SCADA functions stay operational during failures or attacks, thereby eliminating dependence on a single point of failure. Cryptographic hash verification is employed to confirm that software, firmware, and safety-critical packages remain untampered prior to installation, countering vulnerabilities exploited by Stuxnet-style attacks. Strict whitelist-based Access Control Lists (ACLs) at each Demilitarized Zone (DMZ) and network boundary ensure that only explicitly approved devices, protocols, and endpoints communicate across zones, replacing the outdated assumption that internal network entities are inherently trustworthy. Collectively, these mechanisms comply with current guidance, such as NIST SP 800-160 Volume 2, which identifies resiliency techniques, including redundancy, non-persistence, and substantiated integrity, as modern standards. Additionally, NIST SP 800-161 extends this security framework across the supplier base, recognizing that incidents such as Kojima demonstrate that a plant's defenses are only as robust as its weakest connected partner.

Conclusion

This analysis demonstrates how the TMMK architectural design successfully positions security at the forefront of its software delivery and manufacturing monitoring pipeline. As evidenced by the structural layouts detailed within the TMMK Architecture Document, the facility's enforcement of a segmented Purdue Model, firewalled DMZs, and strict multi-layered review sequences establishes a robust framework for operational defense. By isolating critical R&D zones and maintaining stringent physical and digital access controls at crucial interface boundaries, the plant is engineered to mitigate vulnerabilities before they can disrupt production infrastructure. While the systemic reliance on human-carried media across air gaps presents unique boundary challenges, the implementation of localized whitelisted access lists (ACLs) and automated human-machine interface (HMI) log-offs reflects an advanced systems security engineering approach tailored to modern industrial demands.

Ultimately, validating the physical trustworthiness and functional safety of the RAV4 Hybrid depends entirely on the robust engineering controls safeguarding these underlying industrial systems. Because the Distributed Control System (DCS) coordinating robotic assembly, the Programmable Logic Controllers (PLCs) executing mechanical loop logic, and the AGV SCADA networks managing logistics are completely interconnected, the factory floor serves as the definitive root of trust for consumer safety. If a cyber-physical vulnerability allows malicious payloads to alter firmware during assembly, the downstream consequences do not manifest as isolated data leaks, but rather as real-world safety hazards affecting features such as Toyota Safety Sense 2.0 on public roads. Protecting the modern consumer requires continuous vigilance, continuous network monitoring, and an unwavering commitment to defense-in-depth across the entire converged IT-OT manufacturing ecosystem.

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